

Assignment 2 Write Up

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Q1a.

The goal for this dataset is to use it to study trends in stock price fluctuations or trading volumes and find whether large shifts correspond to major company or market events. Additionally, since this will likely be the case, we would like to connect the specific events to these changes if possible. Our group was driven to use this dataset in particular, as we wanted our project to be involved with finance. The mode of collection is observation, as the market changes are observed on each day of the year and stored in a dataset.

Q1b.

The dataset can be accessed through Kaggle for free, which makes the project feasible since it just requires a one-time internet connection to download the data. The data was collected through different measurement processes at stock exchanges. They recorded daily information on trades and compiled it into the following fields: date (specific trade date), open (opening price), high (maximum price during the day), low (minimum price during the day), close (close price adjusted for splits), adj close (adjusted close price adjusted for both dividends and splits), and volume (the number of shares that changed hands during a given day). The dataset includes information on the years of 1997-2020, which is large enough to provide a meaningful analysis.

Q1c.

This project is stored on a Google drive, allowing any viewers or collaborators to download. However, only the team members are allowed to edit. The logical collection for this project will consist of the Amazon daily stock trading dataset, organized as a single dataset with a consistent CSV structure (AMZN.csv). All files have unique and straightforward names following a similar naming convention to clearly distinguish different sets of data. Overall, this structure of organization is accessible and convenient for the team members and is self-documenting and intuitive for viewers or collaborators.

Q1d.

The metadata for the dataset can be accessed through a metadata.yaml file stored in the Google drive. This includes dataset structure, AMZN metadata, provenance, and any metadata from the origin. Additional metadata will be included and show any steps taken in data processing, which will be stored in the same file. Keeping this metadata for collaborators to view is imperative to maintaining transparency and the correct interpretation of the data.

Q1e.

[Link to dataset](#) (The AMZN.csv file in the stocks folder, the metadata is retrieved from the symbols_valid_meta.csv file)

Q2a.

All nine areas of our data management plan worked as originally planned out in assignment one. Interoperability Support, Security Support, Persistence, Data Distribution and Publication, Data Ownership can be highlighted as being extremely effective. "Data Ownership" and "Data Distribution and Publication" were maintained by providing credit to the original author of the dataset in the README.md file present in the Google Drive folder. "Data distribution and Publication", "Security Support", and "Persistence" were maintained by ensuring that the original data is left untouched in our google drive folder. It is also publicly viewable which restricts editing access to prevent unauthorized data tampering. "Security Support" and "Persistence" were further maintained by storing copies of the original dataset in the Google Drive, ensuring that the data has backups available should something happen to the datasets on our local hard drives. "Interoperability Support" was maintained by keeping the data in its original CSV file format. This format is supported on many platforms such as R, Python, and Excel. Therefore, there is little reason to change data formats.

Q2b.

There is not much to change about this fairly straightforward data collection process. However, creating a way to automatically convert the CSV data files into JSON might provide an even wider array of options for working with and sharing the data and further improve the interoperability, especially with web based applications. This data set includes many CSV files, so it is important that the process is programmatic, rather than manual.

Q2c.

Our data management plan outlines how data and metadata will be stored, named, and accessed, making it convenient for the collaborators. And with interoperable file formats such as CSV and YAML, this should enable a more reliable dataset that is easy to analyze. In addition, our data management plan guarantees that metadata is complete, consistent, and meaningful, ensuring transparency in our data quality.

References:

[Amazon Stock Dataset](#)